**SDC HACKATHON 2026**

# Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | HRISSA |
| **Project / Product Name** | Gesir.ai |
| **Final Live Deployment URL** | https://gesir-ai.replit.app (dashboard at /app; create an account at /signup) |
| **Codebase Repository URL** | https://github.com/Azer884/Bridge-app |
| **Name (Leader)** | Azer Hamdaoui |
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## 2. Project Description

***[Maximum 250 words]***

Gesir.ai (called Giser.AI in Phase 1) is a WhatsApp assistant for service SMEs. The name comes from the Arabic word jisr, “bridge”: it connects the customer and the business. It was built first for Malaysian painting contractors and now also serves aircon servicing.

The chore it removes is turning a WhatsApp enquiry into an approved quotation and a booked visit. Gesir chats with the customer in English or Bahasa Malaysia, reads text, photos and voice notes, and asks one question at a time until the job is complete. A deterministic rate engine, never the AI, prices the job from Malaysian market rates. Missing details and unusual cases (damp, mould, a customer who wants a person) go to the owner with a reason.

The owner stays in control. Nothing reaches the customer until a person with the Approver or Admin role approves the estimate in the dashboard or the mobile app; the server enforces the role, a rejection needs a reason and sends nothing, and every step is written to an activity log. The dashboard follows Orbyte’s layout (Pending Actions, Clients, Projects, Users, Wallet) so it can sit inside the Orbyte hub, and a second business or trade is a configuration change, not a rebuild.

Customers agree to a PDPA notice before anything is read, and can erase their data by sending STOP. Gesir cuts the owner’s time on enquiries and helps small teams answer more customers without hiring (SDG 8.2 and 8.3).

## 3. Upgraded Product & Technical Execution

### 3.1 Final System Architecture

*Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.*

Channel. A customer writes to the business's WhatsApp number. The Meta WhatsApp Cloud API delivers each message (text, photo, voice note) to a signed webhook on a FastAPI backend (Python 3.12) hosted on Replit, with PostgreSQL for conversations, quotes, bookings, the audit log and accounts.

Conversation. Code, not the model, runs the flow: PDPA consent first, then the missing job details one at a time, then address, contact, a recap and confirmation. Dates are checked against working hours and public holidays, and addresses against the service area. Qwen2.5-7B-Instruct reads free text and writes replies; Qwen2.5-VL-7B-Instruct reads wall and aircon photos; Whisper (faster-whisper) transcribes voice notes. All run on RunPod serverless GPUs with vLLM. A value a model extracts is kept only if it matches the customer's own words.

Decision. src/pricing.py and src/pricing_aircon.py price from rate tables (config/), so the same job gives the same price. Inspection cases go to a person. The scheduler suggests area-aware slots with travel gaps and never double books.

Approval. The estimate waits in Pending Actions. POST /api/quotes/{id}/approve and /reject check the caller's role on the server; the WhatsApp message is sent only inside the approve call; both outcomes are written to the audit log.

Front ends. A React + TypeScript + Vite dashboard in Orbyte's layout (served at /app after sign-up or sign-in) and an Expo React Native app with push notifications for the owner. Signed events (HMAC-SHA256) let an Orbyte hub follow quotes. Experimental: WhatsApp calls through LiveKit (voice_worker/).

### 3.2 Implementation & Hackathon Deliverables

*Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.*

***[Maximum 300 words]***

Built and deployed at gesir-ai.replit.app (repository history from 21 Oct to 1 Nov 2026, more than 140 commits): the full journey from a WhatsApp message to an approved, sent estimate and a booked visit, in English and Bahasa Malaysia, with text, photos and voice notes.

Changes since the Phase 1 design: (1) We dropped Vapi and Google Calendar. Our own scheduler books slots with travel time, holidays and no double booking, and LiveKit replaced Vapi for experimental calls. (2) Approval is enforced in the backend with three roles (Admin, Approver, Reviewer); a Reviewer gets 403. Rejection needs a reason, sends nothing to the customer and is logged. (3) PDPA consent before any message is read, STOP to erase, owner erasure. (4) A second trade, aircon servicing, with its own rules file and rate card, proving that a new business is configuration. (5) The dashboard was rebuilt in Orbyte's layout with Wallet credits and signed hub events. (6) A new mobile app with an Approve tab and notifications. (7) Real sign-up and sign-in, where the owner picks painting or aircon; both trades work for every account for now.

Problem-solving: GPU workers take minutes to wake, so we added waiting messages, a status page and retries. A 7B model misread messy messages, so every extracted value is checked against the customer's words and anything unclear is asked again or handed to the owner. We ran about 200 stress conversations against the live model, read every reply and added regression tests. The automated suite now has 607 tests, including that the same input gives the same price.

## 4. Environmental & SDG Impact Verification

*Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.*

*The data and evidence you provide in this section will be used to assess* ***SDG Impact, Technical Execution and Innovation scores.*** *Vague claims or missing verification will weaken your score across all three categories.*

| **Metric Category** | **Your Response** |
|---|---|
| **Target SDG Metric Achievement**<br>*Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* | Modelled, not yet measured with a live SME. Baseline from Phase 1: 12.5 owner hours a week (midpoint of 10–15). Assuming 20 enquiries a week, that is 37.5 minutes of owner time per enquiry. With Gesir it is 3.25 minutes: 2 minutes to review, 0.5 to approve, and 15% of chats handed to the owner at 5 minutes each. Result: |

| Metric Category | Your Response |
|---|---|
| *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."* | 11.4 hours saved a week, a 91.3% reduction against the 85% Phase 1 target, and manual steps per enquiry cut from 13 to about 2.<br><br>The target holds up to about 34 enquiries a week (82.7% at 40, 95.7% at 10). Running cost is an estimate of US$0.01–0.02 of GPU time per quote at RunPod's listed rates. The Phase 1 benchmark of $3–$5 per chat interaction (Ringly.io, 2026) is an outside figure, not our measurement. |
| **Verification Evidence Method**<br>*How was this data verified?*<br>*e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.* | The model is reproducible: python scripts/benefit_model.py prints every number above and takes each assumption as a flag. The 13 baseline steps are the 9 fields in config/business_config.example.json plus working out the area, pricing, writing the quote and offering a slot.<br><br>To replace the assumed minutes with measured ones, every message, estimate, approval, rejection and booking is time-stamped in the event log (/api/events) and the audit log (/api/audit). Correctness is checked by 607 automated tests (same input, same price) and about 200 stress conversations (scripts/stress_chat.py) run against the live model. |
| **Socio-Economic / Civic Value**<br>*How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?* | Gesir raises the productivity of micro and small businesses through technology (SDG 8.2) and lets a small team answer more customers without hiring (SDG 8.3). It fits Orbyte as a shared business hub, so one integration reaches many service SMEs: painters, aircon technicians and then plumbers, electricians and cleaners. SME-support bodies such as SME Corp Malaysia could back pilots as part of SME digitalisation. |
| **Deployment & Commercial Feasibility**<br>*e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.* | B2B SaaS on the WhatsApp Business Cloud API, sold through the Orbyte hub. Pricing is usage-based, one wallet credit per estimate a person approves, and nothing for chats that never become an estimate. Hosting is a Replit deployment with PostgreSQL; models run on RunPod serverless GPUs that scale to zero. A new business is a config file and a rate card, not new code. |

### 4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

*Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.*

*This proof is graded across three categories at once:* ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

```
# scripts/benefit_model.py (core)
before_min = baseline_hours * 60 / enquiries
with_min   = review_min + approve_min + handover_rate * handover_min
reduction  = 100 * (1 - with_min / before_min)
saved_h    = enquiries * (before_min - with_min) / 60
steps_before = len(required_fields) + 4     # 9 fields + area, price, quote, slot
steps_with   = 2 + handover_rate            # review, approve, + replies

$ python scripts/benefit_model.py
owner_minutes_per_enquiry_before           37.5
owner_minutes_per_enquiry_with_gesir       3.25
owner_hours_per_week_with_gesir            1.08
hours_saved_per_week                       11.4
time_reduction_percent                     91.3
manual_steps_before                        13
manual_steps_with_gesir                    2.15
steps_removed_percent                      83.5

Check: 37.5 min x 15% = 5.6 min is the most the owner may spend per enquiry to reach
85%; 3.25 < 5.6.
Break-even: 85% holds while enquiries <= 750 x 0.15 / 3.25 = 34.6 a week.
```

## 5. Market Viability & Future Scalability Plan

*SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?*

***[Maximum 250 words]***

Pilot first. We would run Gesir with about ten painting and aircon businesses in Selangor, introduced through XMUM and the Rotary and Rotaract networks, to replace our assumed minutes with measured ones from the audit log.

Make it production-ready: a verified WhatsApp Business number with Meta-approved templates, so estimates can be sent after the 24-hour window; real per-person accounts and roles instead of the simulated Users list; automatic data expiry and customer data export for PDPA; database backups, alerting and CI.

Scale the product: each business is a config file plus a rate card, so onboarding is a setup task. We would add plumbing, electrical and cleaning, each with a rules file and a rate card supplied by a real contractor. GPU use is billed per second and scales to zero, so cost grows with quotes sent, not with sign-ups.

Earn money: sell through Orbyte on its wallet, one credit per approved estimate. Funding would pay for the GPU, onboarding and support during the pilot. The price per credit is a proposal to test in the pilot, not a measured figure.

Grow: Malay and English are done; Mandarin and Tamil are the next languages for Malaysia’s SMEs. Success is judged on measured owner hours saved and quotes approved per business.

## 6. Final Submission Checklist

*Tick all boxes that apply before submitting.*

- [x] The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- [x] The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- [x] The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- [ ] The final pitch deck has been exported and saved as a flattened PDF file.
- [x] All fields in this template have been completed and word limits respected.

## 7. IP & Originality Declaration

### 7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

### 7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

| Name | Type | License |
|---|---|---|
| Qwen2.5-7B-Instruct | Chat model (RunPod, vLLM) | Apache-2.0 |
| Qwen2.5-VL-7B-Instruct | Photo model (RunPod, vLLM) | Apache-2.0 |
| faster-whisper (SYSTRAN) | Speech recognition code | MIT |
| Whisper small and base | Speech recognition model | MIT |
| FastAPI, Uvicorn, Pydantic | API framework and server | MIT, BSD-3-Clause |
| vLLM (RunPod worker) | Model serving | Apache-2.0 |
| LiveKit Agents and Cloud; Deepgram Nova-3, Cartesia Sonic-3.5 (via LiveKit Inference) | Voice calls (experimental) | Apache-2.0; provider terms |
| React, Vite, TypeScript, Expo / React Native | Dashboard and mobile app | MIT, Apache-2.0 |
| PostgreSQL driver psycopg 3 | Database access | LGPL-3.0 |

### 7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

## 8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



1 November 2026